

Pearl Natalia

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Education

University of Waterloo

Sept 2023 - Apr 2028

Bachelor of Software Engineering (Honours), Co-op

Waterloo, ON

- **Involvement:** Women in Engineering (Sponsorship Director)
- **International Exchange:** Tecun School of Engineering, University of Navarra, Spain (Jan–May 2026)

Technical Skills

Languages: Python, Java, SQL (Postgres), C, C#, C++, JavaScript, TypeScript, HTML/CSS
Tools: AWS, MongoDB, Snowflake, Databricks, Neo4j, Docker, Google Cloud, Modal, Jenkins, Trino, Terraform, Amplitude
Frameworks: React, Angular, Flask, Node.js, Next.js, Django, Bootstrap, Spring Boot, ROS
ML Libraries: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, OpenCV, Matplotlib, Keras, MLflow, LangChain

Experience

AI Engineering Intern

June 2026 – Present

Carebrain ↗

San Francisco, CA

- Built an automated clinician scheduling system using LLM-structured output parsing and a greedy optimization algorithm
- Improved speaker diarization accuracy using a two-phase LLM-based correction pipeline with role-aware context handling

Machine Learning Intern

Sept 2025 – Dec 2025

The Globe and Mail ↗

Toronto, ON

- Built a deep-learning retention-offer model by feature engineering Amplitude data, projecting **\$300k/year** in churn savings
- Engineered a Slackbot (Databricks, Anthropic, AWS) to surface and send journalists personalized briefings, **saving 3h/week**
- Productionized a search API (FastAPI, Terraform, ECS), **decoupling CPU/GPU** workloads to optimize inference and load
- Led end-to-end projects with Legal, Sales, and Editorial stakeholders to design solutions that supported key team objectives

Machine Learning Engineer

Apr 2025 – May 2025

Legacy ↗ · Contract

San Francisco, CA

- Created a responsive **React web app** with **SQLite** for fast data retrieval, and **hosted ML models** on Modal
- Designed a **RAG pipeline** to generate accurate counselor advice by retrieving similar patient cases from a vector database
- Trained an anxiety prediction model with **Naive Bayes**, **Random Forest classifiers** and **TF-IDF** vectorization
- Built a **semantic search** over **14k** counseling dialogues using embeddings and optimized results via LLM-based preprocessing

Software Engineering Intern

Jan 2025 – Apr 2025

PointClickCare ↗

Toronto, ON

- Built a **Grafana** dashboard with **Docker** to visualize **Trino** pipeline metrics, generating **100k datapoints** for **stress testing**
- Developed a Spring Boot tool to convert SQL queries to Trino format via syntax trees, enabling **scalable** federated querying
- Collaboratively optimized batch sizes to reduce latency in migrating **millions** of health records across **distributed databases**
- Documented iterative progress in **Confluence** following **Agile methodologies** across all projects

Web Developer Intern

May 2024 – Aug 2024

Generis Global Partners ↗

Toronto, ON

- Automated data cleansing via a web app (CLIP embeddings, web scraping) hosted on **AWS EC2**, saving **20h/week**
- Rebuilt 15 websites to **increase traffic and SEO by 35%** using **Google Analytics** and marketing team mockups
- Built a Chrome extension to parse emails via **Flask** for new website content, **accelerating development time by 73%**

Projects

Agentic Macbook Voice Assistant ↗ | *SQLite, Whisper API, Gemini Flash, Spotify API*

- Created an agentic MacBook voice assistant to control system settings and calendar events, play music, write emails and more
- Built **chain-of-thought reasoning** with Gemini and **NLP** to convert user requests into executable AppleScript commands
- Generated iMessage replies via few-shot prompting and tracked high-frequency phrases using **SQLite** to replicate user style

AI Wearable for SOS (HackThe6ix) ↗ | *React, Flask, Auth0, MongoDB, Gemini Flash, Hume AI, Mapbox*

- Led the hardware integration of a SOS device to capture video and audio recording with a Raspberry Pi and Flask
- Generated **contextual summaries of video frames** via **Gemini Flash** and **Hume AI APIs**, highlighting critical threats
- Developed a **CNN-based audio classification model** with spectrograms using TensorFlow to classify violent audio clips

AI-Powered Dash Cam ↗ | *OpenCV, Numpy, HERE Maps, Pandas, Scikit-Learn, React, SQLite*

- Developed **low-latency geolocation algorithms** to detect vehicle speed/turns, road violations and speed cameras
- Implemented real-time traffic light and road sign detection from live dashcam footage using YOLOv8, Flask and FFmpeg
- Improved **CNN model** accuracy on low-quality frames via relative positioning, color masking and Hough transforms